

**ECE 3337: Signals & Systems Analysis**

**Lecture 17: System Synthesis**

**Reference: Chapter 5  
(Section 4.7, 5-1)**

**Please turn off your gadgets & other distractions now**



# Exam 2

---

**Syllabus: Chapters 1 - 4, as covered in class**

**Time: Saturday, April 4<sup>th</sup> , 10 am – 11:30 am**

**Location: Science & Research Building room 117 (both sections)**

## **Help Sessions**

**March 27<sup>th</sup> 10am – noon, in-person @ Cemo Hall Room 109 (Dr. Roysam)**

**April 3<sup>rd</sup> noon – 2pm, Online via MS-TEAMS (Dr. Reddy)**

**Old Exams with solutions are on the Canvas site for practice**

- 1. Try the old exams individually without looking at the answers**
- 2. Attempt the problems and practice until you can do them fully on your own without notes, quickly, just like in a test.**
- 3. Practice for correctness and speed**



# How to Build Systems!

Thus far, we have focused on analyzing a given system

**We now consider the reverse problem: building a new one!**

- Devise a system that realizes a given transfer function  $H(s)$
- For example, given,

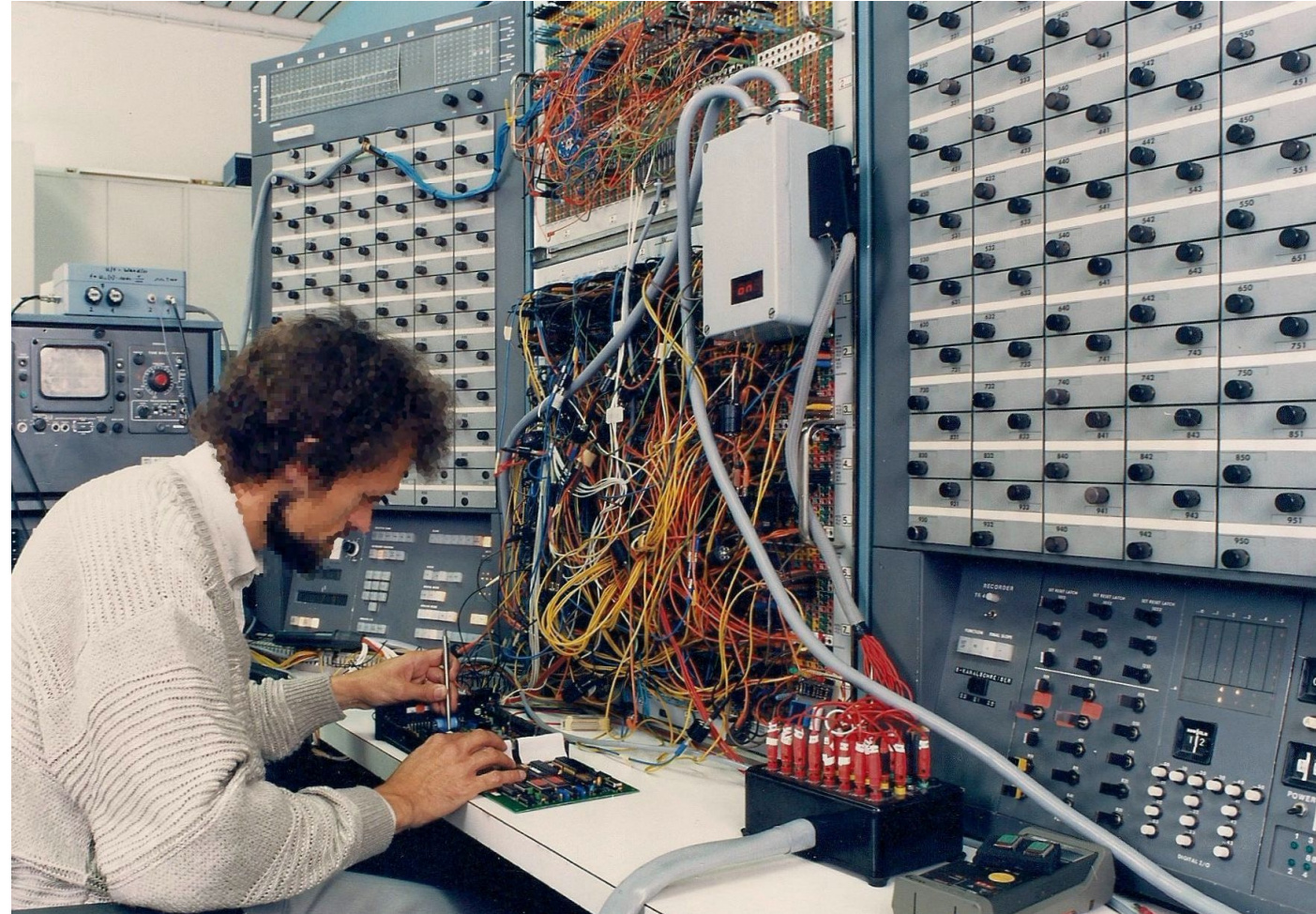
$$H(s) = \frac{b_0s^3 + b_1s^2 + b_2s}{s^3 + a_1s^2 + a_2s + a_3}$$

- We want to create a system in the form of a block diagram consisting of adders, integrators, and inverters to realize  $H(s)$
- Integrators are preferred over differentiators since they are stable, precise, and easier to work with



# Related History: Analog Computing

- In the 1960's analog computers were widely used to simulate LTI systems
  - They were faster and cheaper than the digital computers of the era
- A general-purpose analog computer was equipped with a “wall” of op amps, resistors, capacitors, inductors, etc.
- Users “programmed” the analog computer by connecting wires between these generic devices to build a system



By SchmiAlf - Own work, CC BY-SA 4.0,  
<https://commons.wikimedia.org/w/index.php?curid=81886020>





# Analog computing is useful for AI



<https://mythic.ai/>

ENTERPRISE TECH

## Mythic Launches Industry First Analog AI Chip

Karl Freund Contributor

Founder and Principal Analyst, Cambrian-AI Research LLC

Follow

Jun 7, 2021, 09:00am EDT



Listen to article 5 minutes



### Mythic claims 400 TOPS with 1.28B parameters on 16-chip PCIe Card

*Please welcome new Cambrian-AI Analyst Gary Fritz, who contributed to this article.*

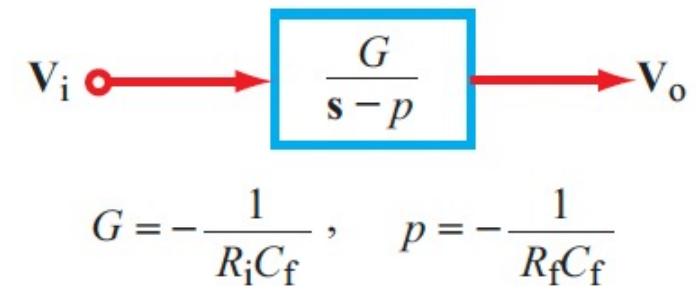
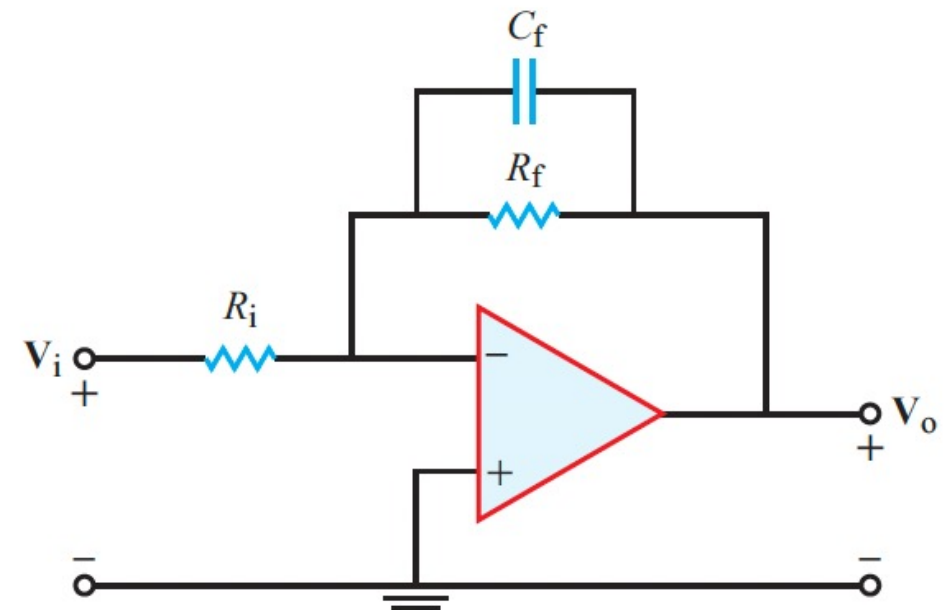
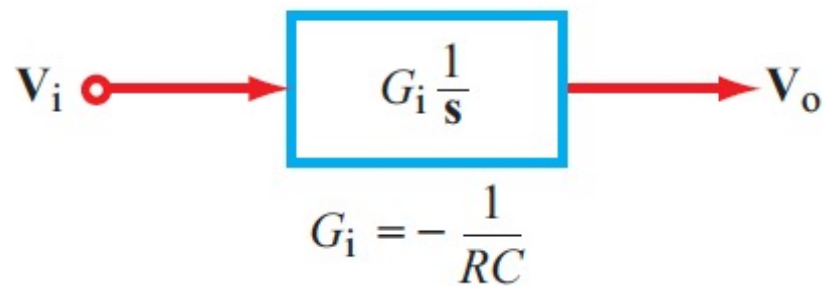
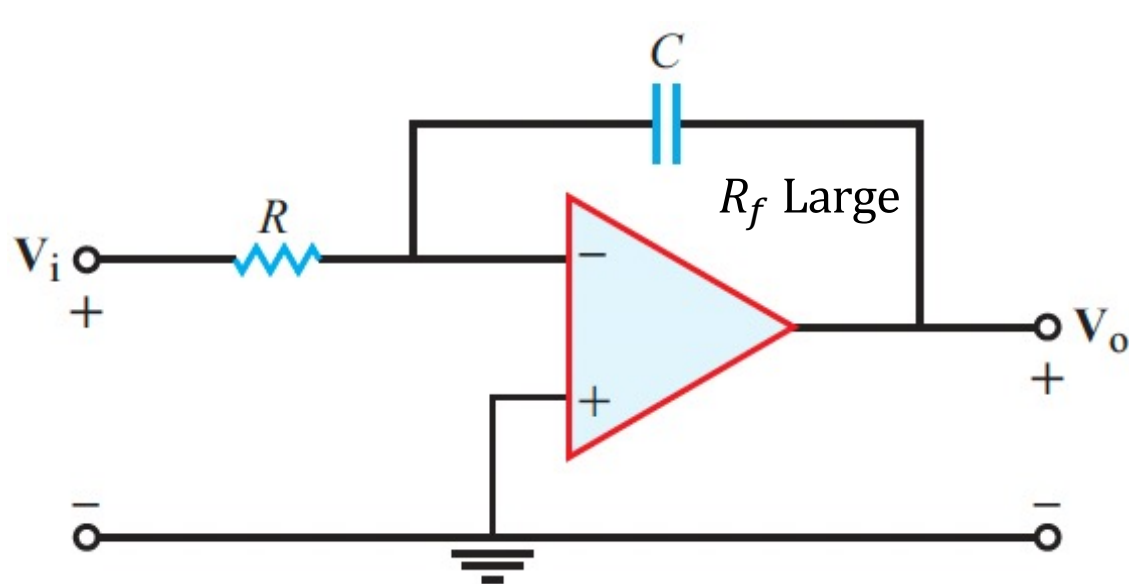
Artificial Intelligence applications are starting to show up in everything from cell phones to supertankers. But at the edge, they are running into the same roadblocks that traditional applications have fought for years: they need more speed. More capacity. Lower power. What's a burgeoning neural net to do? To make matters worse, machine learning models are growing at an exponential rate, doubling in size every 3.5 months. In these



# Overarching Principle for Synthesis

We rely on integrators as the basic building blocks and avoid differentiators as much as possible

- Integrators are stable and do not amplify noise as differentiators can



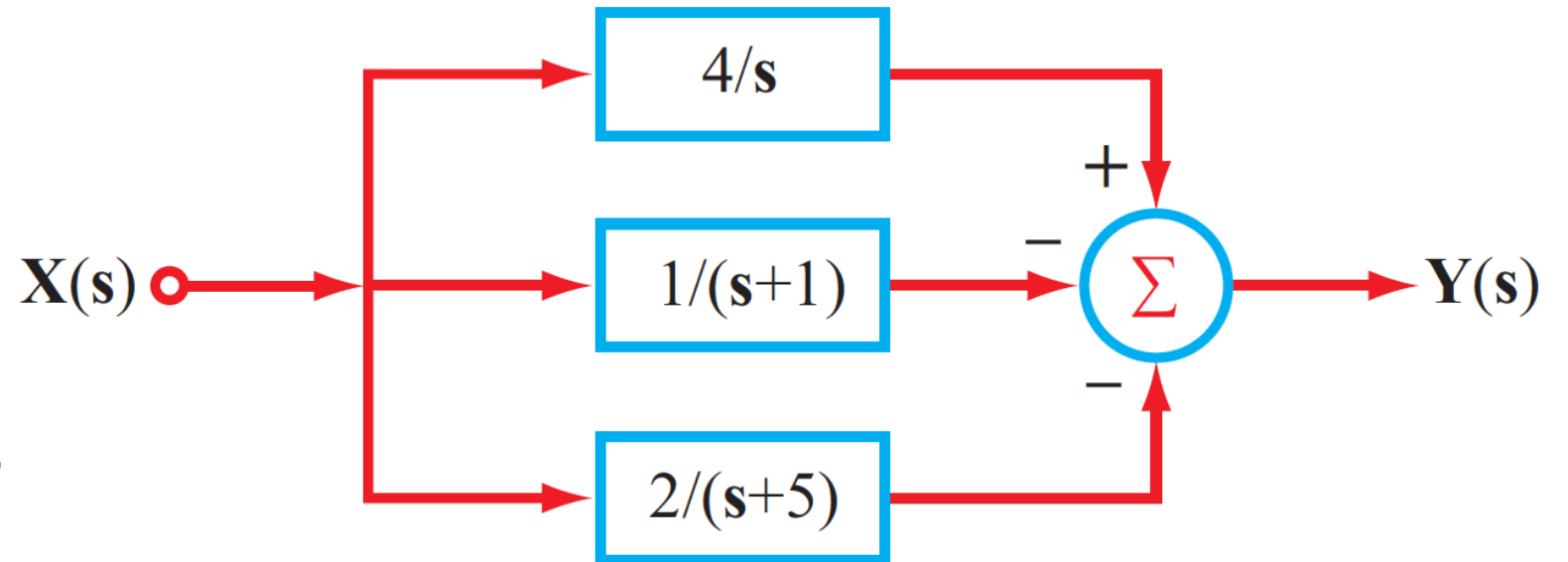


# Method #1: Use Partial Fractions

Partial fraction expansions offer a simple approach for system synthesis.

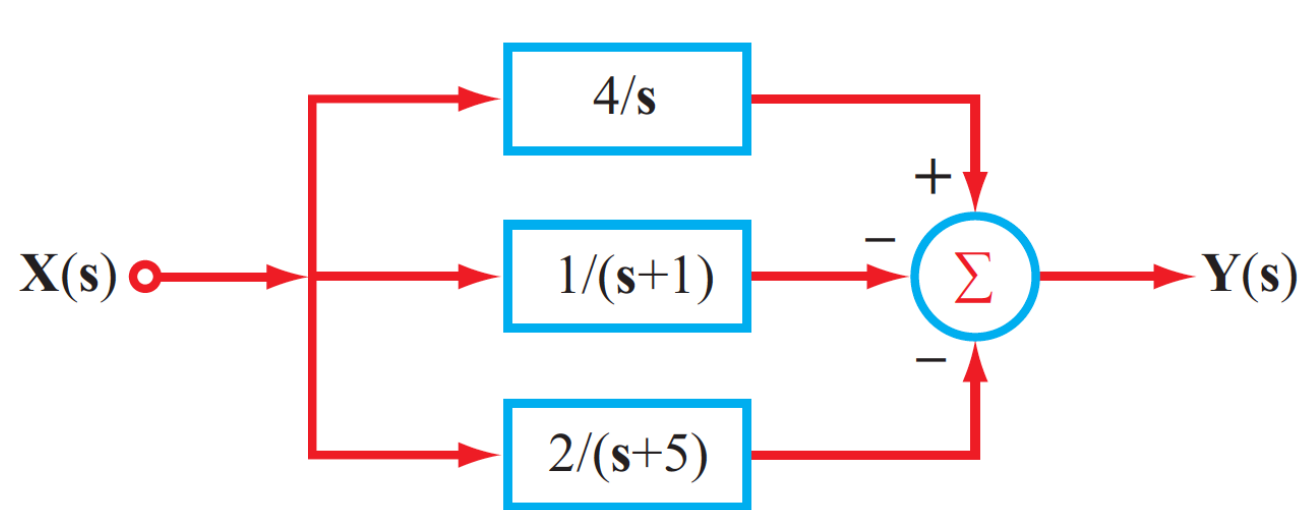
For Example:

$$\begin{aligned} \mathbf{H(s)} &= \frac{s^2 + 17s + 20}{s(s^2 + 6s + 5)} \\ &= \frac{4}{s} - \frac{1}{s+1} - \frac{2}{s+5} \end{aligned}$$

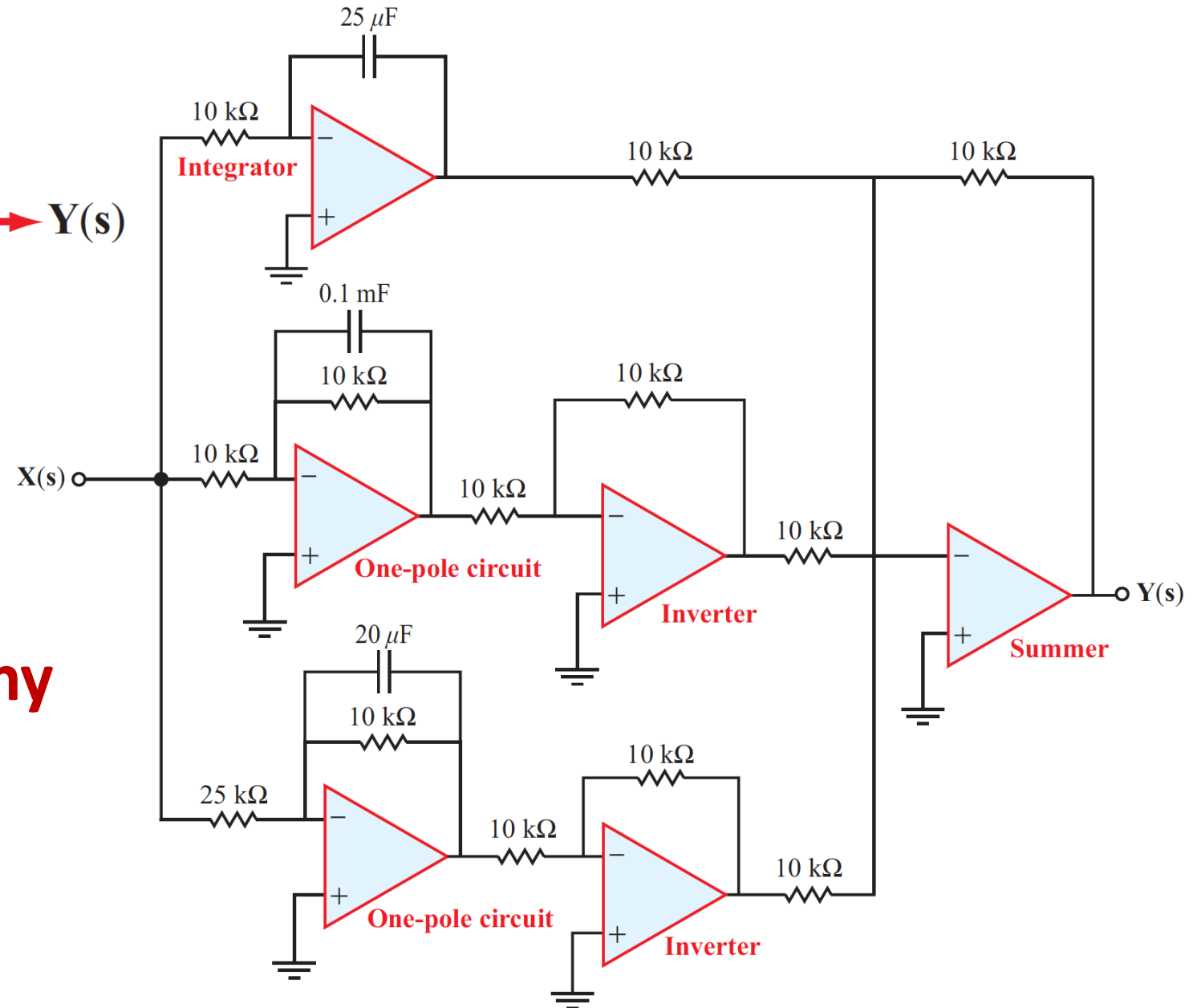




# OP-AMP Implementation



**This approach is easy to understand but uses too many op amps !**

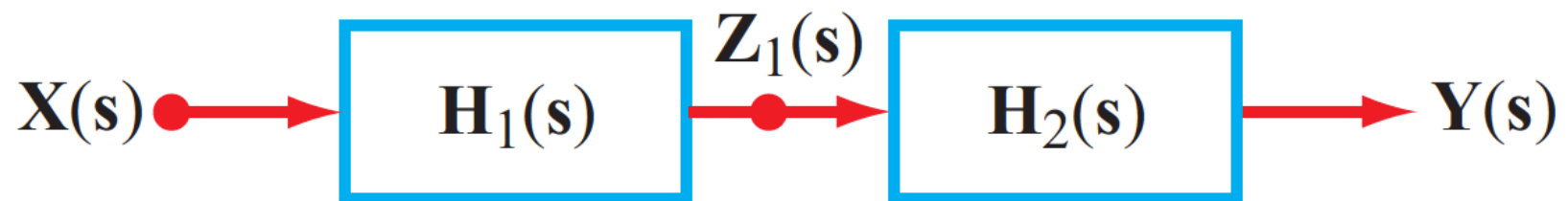






# Method #2: Rewrite the Transfer Function

$$\begin{aligned}\mathbf{H(s)} &= \frac{b_0s^3 + b_1s^2 + b_2s}{s^3 + a_1s^2 + a_2s + a_3} \cdot \frac{1/s^3}{1/s^3} \\ &= \left( b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} \right) \left( \frac{1}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}} \right) \\ &= \mathbf{H_1(s)} \mathbf{H_2(s)},\end{aligned}$$

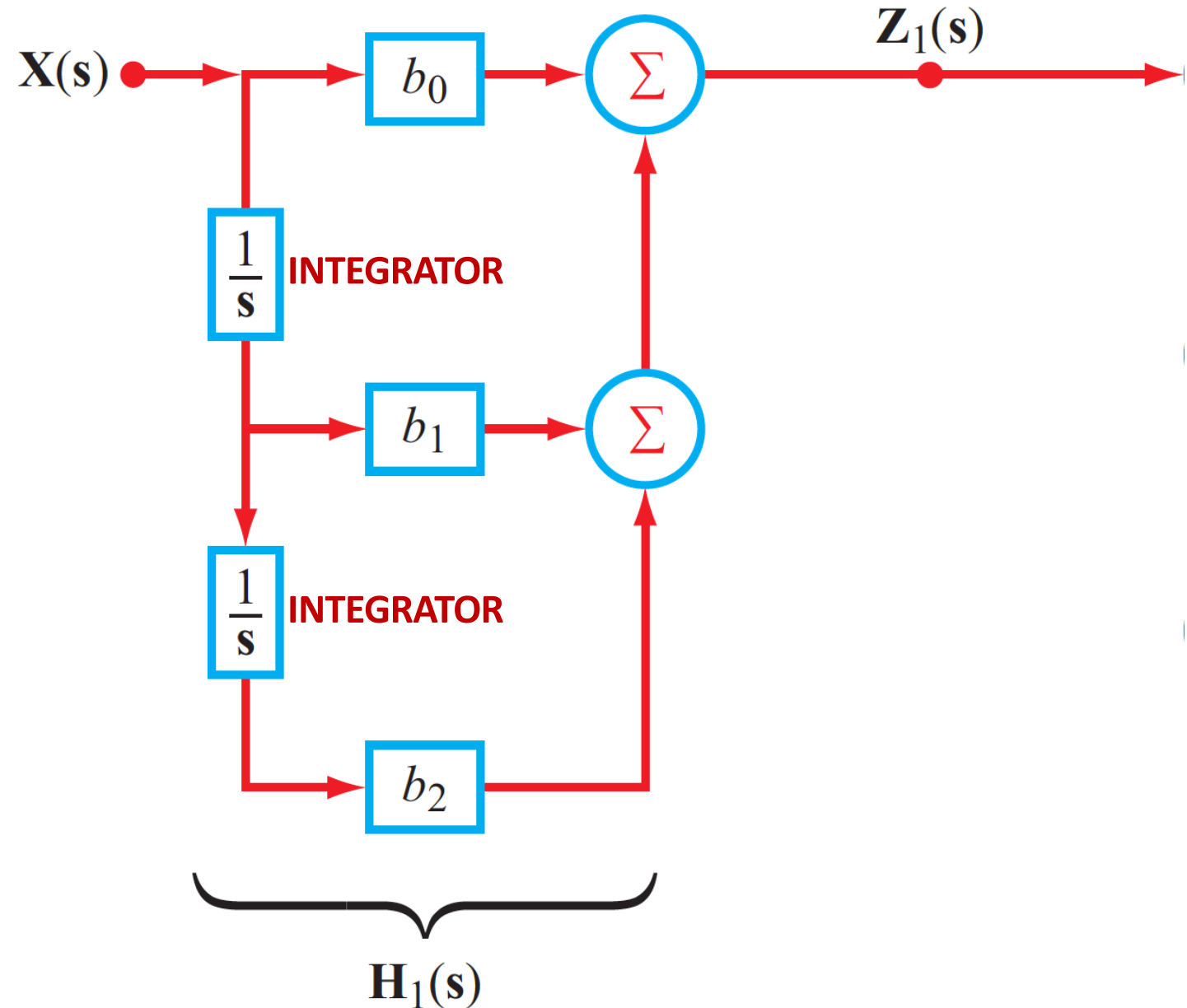




# The First Part

$$\mathbf{H}_1(s) = b_0 + \frac{b_1}{s} + \frac{b_2}{s^2}$$

Note the systematic reuse  
of the op amps to calculate  
higher-order terms





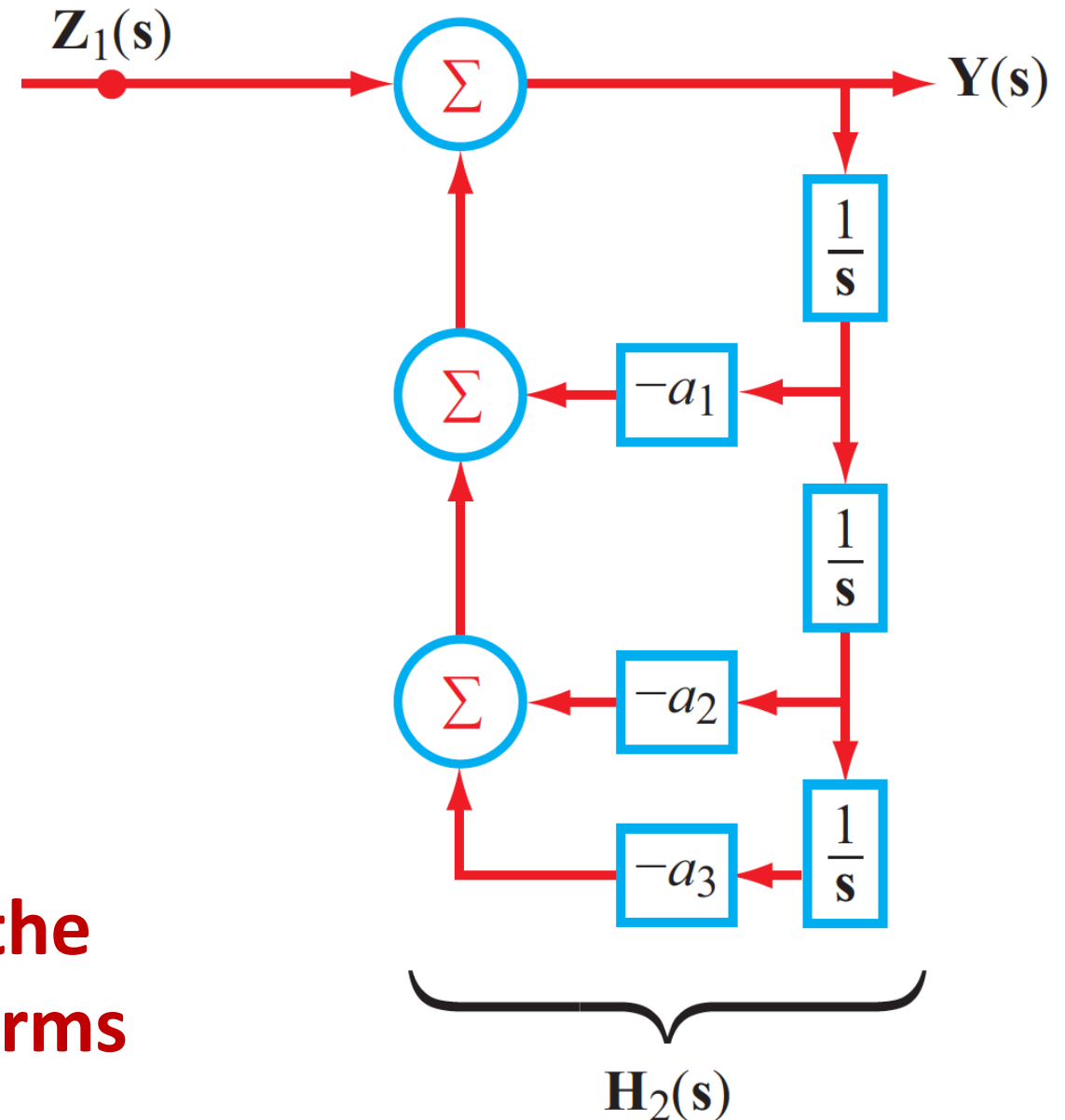
## The Second Part

$$\mathbf{H}_2(s) = \left(1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}\right)^{-1}$$

$$\mathbf{Z}_1(s) = \frac{\mathbf{Y}(s)}{\mathbf{H}_2(s)} = \left(1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}\right) \mathbf{Y}(s)$$

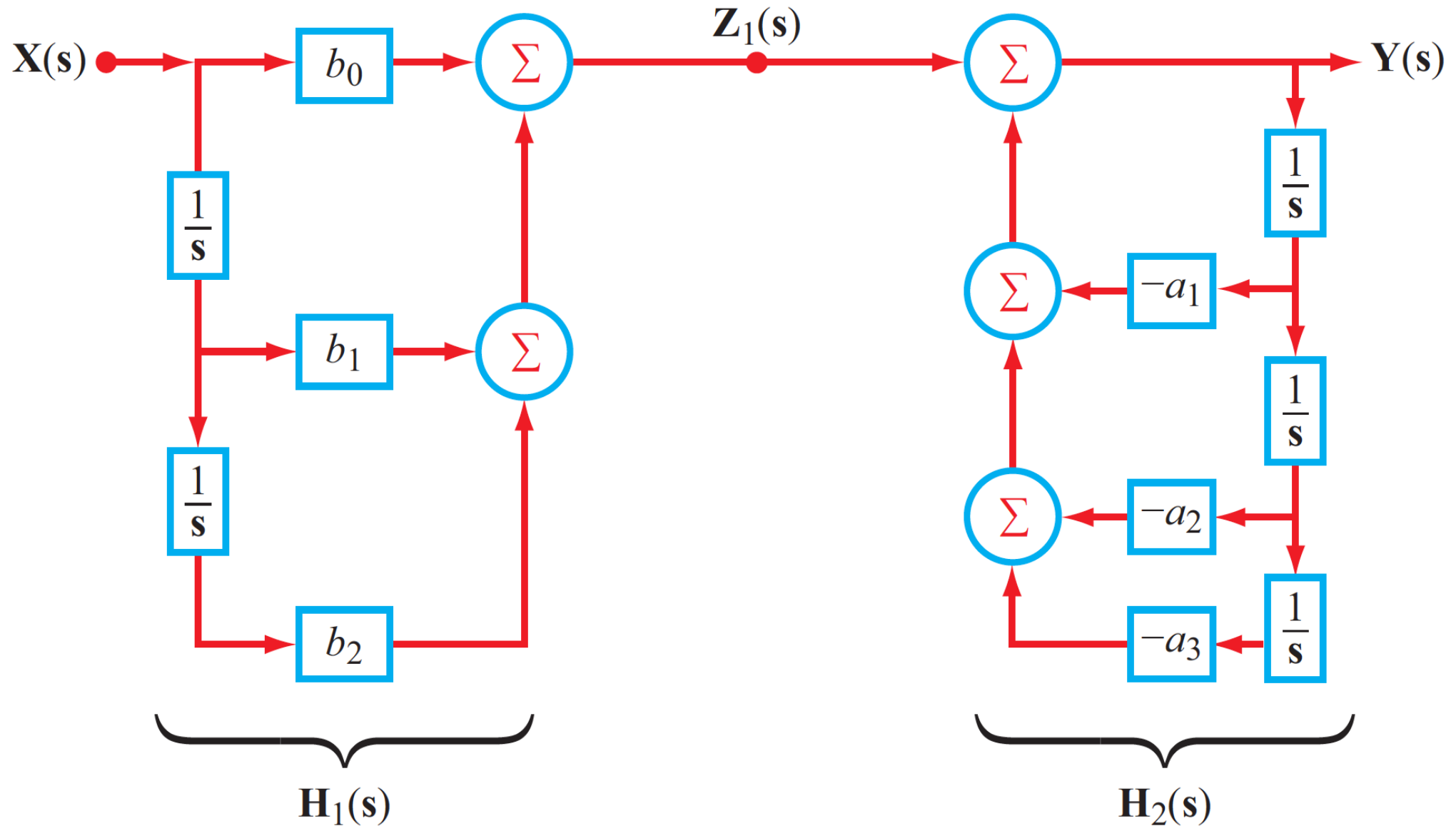
$$\mathbf{Y}(s) = \mathbf{Z}_1(s) - \left(\frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}\right) \mathbf{Y}(s)$$

**Again, note the systematic reuse of the op amps to calculate higher-order terms**





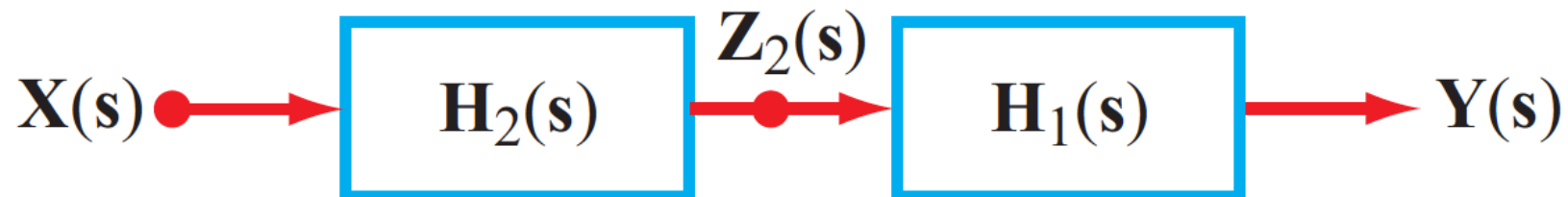
# Put Them Together: Direct Form I





## Another Possibility

$$\begin{aligned}\mathbf{H}(s) &= \frac{b_0 s^3 + b_1 s^2 + b_2 s}{s^3 + a_1 s^2 + a_2 s + a_3} \cdot \frac{1/s^3}{1/s^3} \\ &= \left( b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} \right) \left( \frac{1}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}} \right) \\ &= \mathbf{H}_1(s) \mathbf{H}_2(s),\end{aligned}$$



**SWAP USING COMMUTATIVITY**





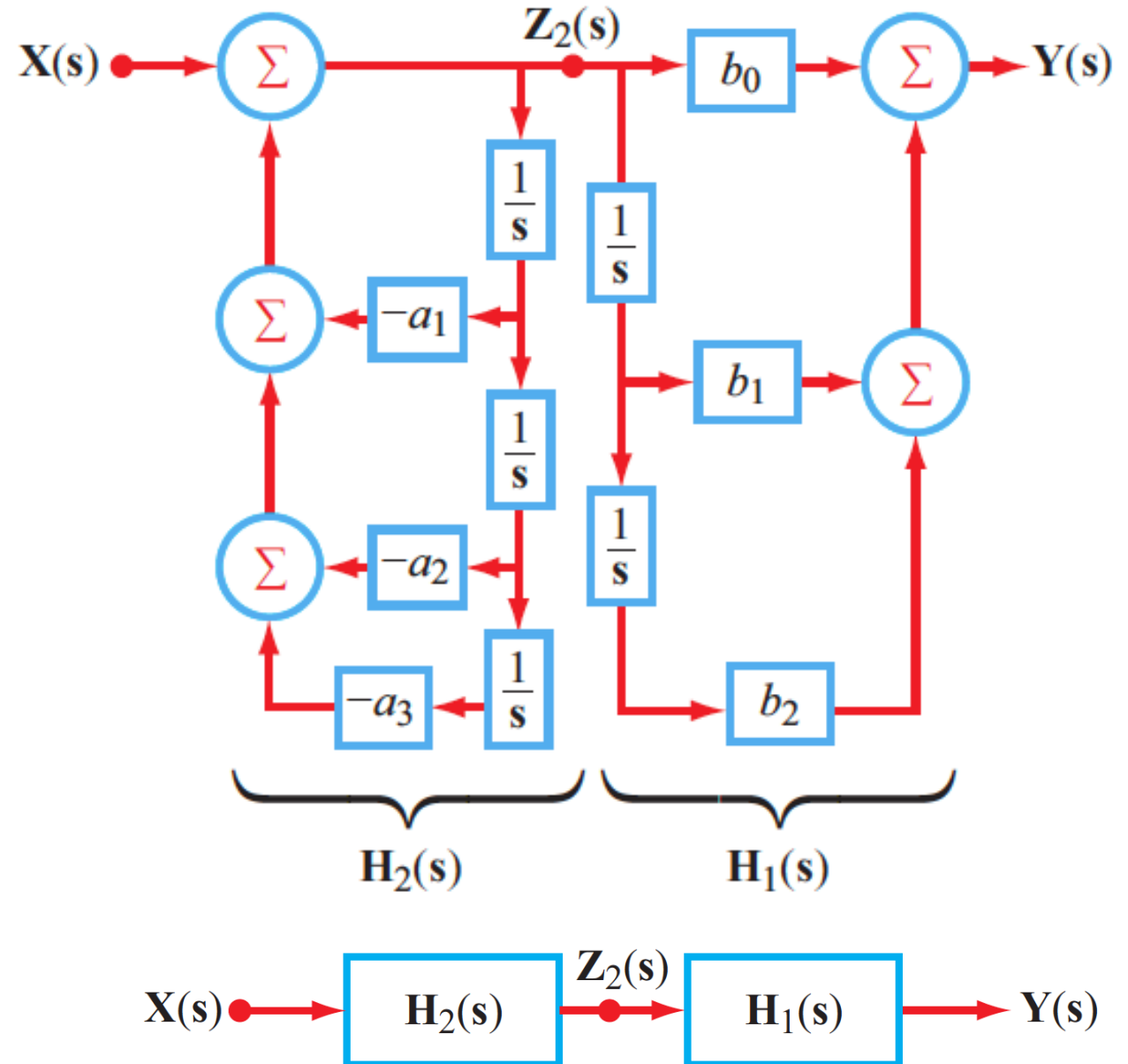
# This Version Offers a New Possibility

$$\begin{aligned} \mathbf{H}(s) &= \frac{b_0 s^3 + b_1 s^2 + b_2 s}{s^3 + a_1 s^2 + a_2 s + a_3} \cdot \frac{1/s^3}{1/s^3} \\ &= \left( b_0 + \frac{b_1}{s} + \frac{b_2}{s^2} \right) \left( \frac{1}{1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3}} \right) \\ &= \mathbf{H}_1(s) \mathbf{H}_2(s), \end{aligned}$$

**SAME AS BEFORE**

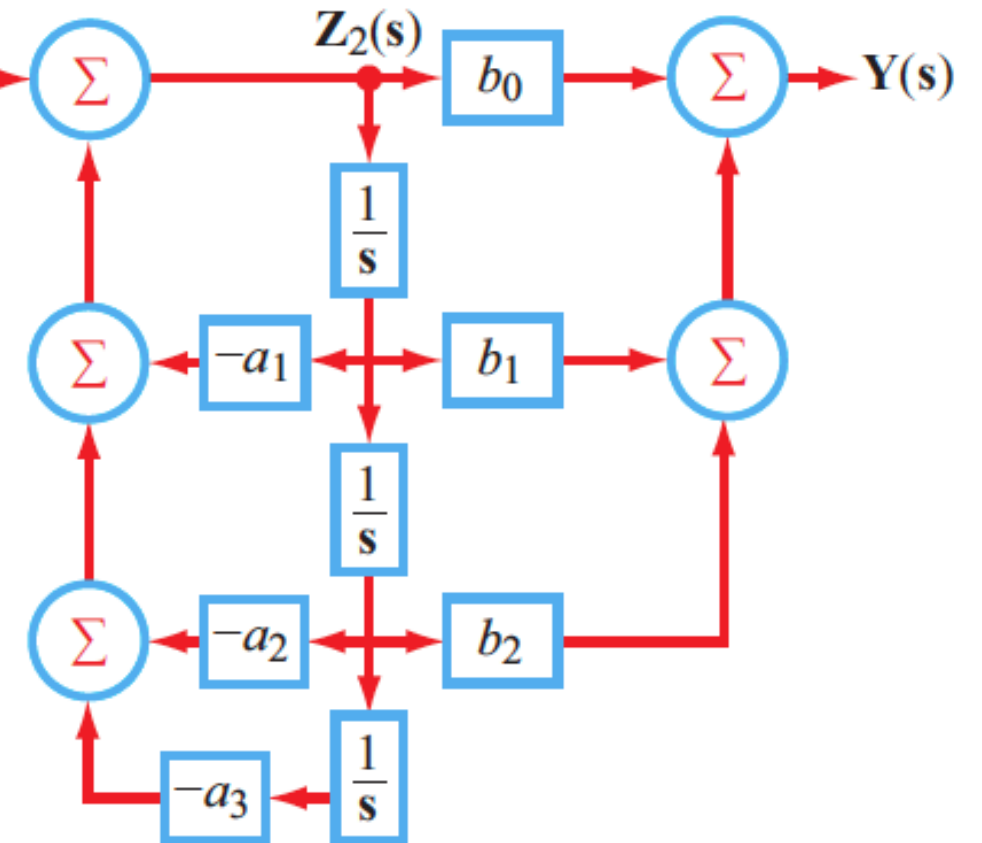
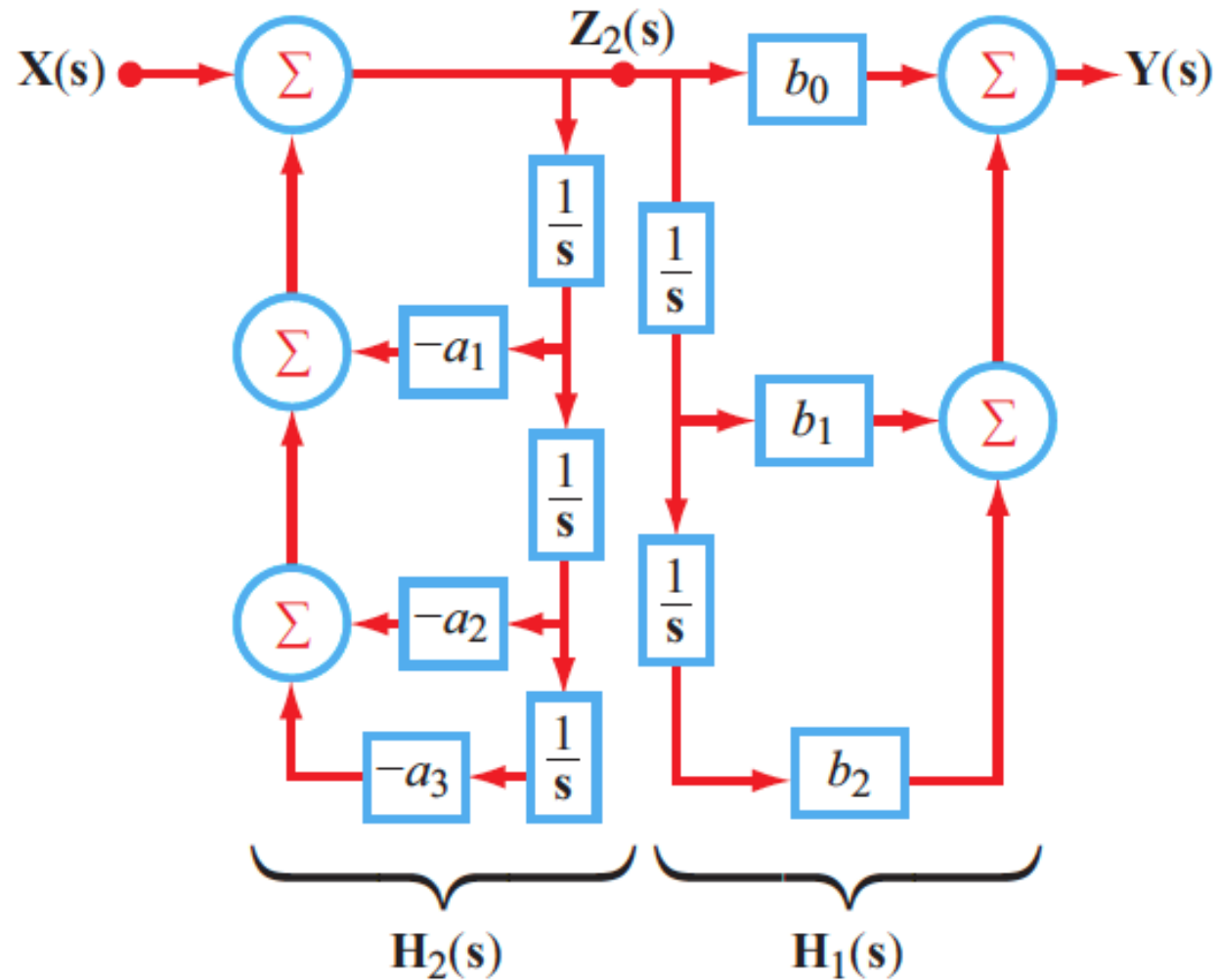
$$\mathbf{H}_1(s) = b_0 + \frac{b_1}{s} + \frac{b_2}{s^2}$$

$$\mathbf{H}_2(s) = \left( 1 + \frac{a_1}{s} + \frac{a_2}{s^2} + \frac{a_3}{s^3} \right)^{-1}$$





# Rewrite Using Shared Integrators: Direct Form II



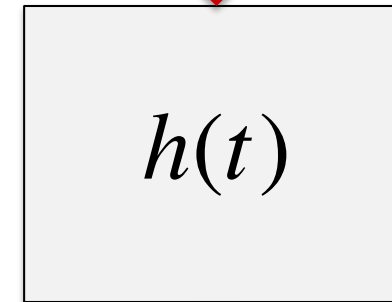
**MUCH MORE COMPACT  
& FEWER OP AMPS !!**



# Recap: LTI Response to Sinusoids

**SINUSOID IN:**  $x(t) = A \cos(\omega t + \phi)$

$$\hat{H}(\omega) = \int_{-\infty}^{\infty} h(\tau) e^{-j\omega\tau} d\tau$$



**SAME FREQUENCY  
SINUSOID OUT,  
CHANGED AMPLITUDE  
CHANGED PHASE**

$$y(t) = A \underbrace{|\hat{H}(\omega)|}_{\text{Amplitude Change}} \cos(\omega t + \phi + \underbrace{\theta}_{\text{Phase Change}})$$

$$\theta = \angle \hat{H}(\omega)$$



# Circuits that Operate at a Fixed Frequency: Phasors

Phasors are a compact way to describe signals at a fixed and known frequency

Given a signal:

$$v(t) = V_0 \cos(\omega t + \phi)$$

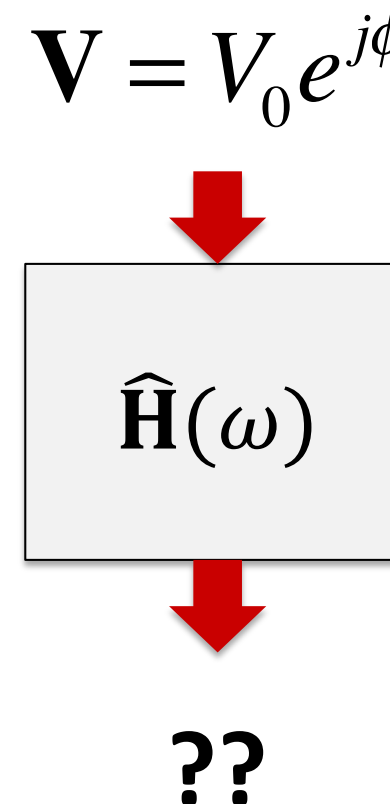
We define the complex number

$$\mathbf{V} = V_0 e^{j\phi}$$

It is the **phasor** associated with  $v(t)$

**Note:** The frequency  $\omega$  is not explicitly written

– it is implicit i.e., understood so not stated explicitly





# Cosine-reference Phasors

Notice that we defined the phase angle relative to the cosine function

- For this reason, we call it the **cosine-reference phasor**
- Cosine is preferred since it is the real part

**For example, consider the following signal:**

$$\begin{aligned}v(t) &= V_0 \sin \omega t \\ &= V_0 \cos(\omega t - \pi / 2)\end{aligned}$$

**Phasor representation:**  $\mathbf{V} = V_0 e^{-j\pi/2}$



## Phasors (contd.)

In general, a sinusoidal signal can be written as follows:

$$x(t) = \text{Re}[\underline{\mathbf{X}} e^{j\omega t}]$$

**PHASOR**

**Observations:**

- The phasor is a complex number with a magnitude and phase:

$$\mathbf{X} = |\mathbf{X}| e^{j\phi}$$

$$x(t) = \text{Re}[\mathbf{X} e^{j\omega t}]$$

$$= \text{Re}[|\mathbf{X}| e^{j\phi} e^{j\omega t}]$$

$$= \text{Re}[|\mathbf{X}| e^{j(\omega t + \phi)}]$$

$$= |\mathbf{X}| \cos(\omega t + \phi)$$

**Observe that the phasor is time independent**





# Phasors and Signal Differentiation

$$x(t) = \text{Re}[\mathbf{X} e^{j\omega t}] = \text{Re}[|\mathbf{X}| e^{j\phi} e^{j\omega t}] \Leftrightarrow \mathbf{X}$$

$$\frac{dx}{dt} = \frac{d}{dt} \left( \text{Re}[|\mathbf{X}| e^{j\phi} e^{j\omega t}] \right) \quad \text{CORRESPONDS TO}$$

$$= \text{Re}[|\mathbf{X}| e^{j\phi} \frac{d}{dt} e^{j\omega t}]$$

$$= \text{Re}[|\mathbf{X}| e^{j\phi} j\omega e^{j\omega t}]$$

$$= \text{Re}[j\omega |\mathbf{X}| e^{j\phi} e^{j\omega t}] \Leftrightarrow j\omega \mathbf{X}$$

CORRESPONDS TO



# Phasors and Signal Integration

Similarly,

$$x(t) = \text{Re}[\mathbf{X} e^{j\omega t}] = \text{Re}[|\mathbf{X}| e^{j\phi} e^{j\omega t}] \Leftrightarrow \mathbf{X}$$

$$\int x(\tau) d\tau = \int \left( \text{Re}[|\mathbf{X}| e^{j\phi} e^{j\omega\tau}] \right) d\tau \quad \text{CORRESPONDS TO}$$

$$= \text{Re}[|\mathbf{X}| e^{j\phi} \int e^{j\omega\tau} d\tau]$$

$$= \text{Re}[|\mathbf{X}| e^{j\phi} \frac{1}{j\omega} e^{j\omega t}]$$

$$= \text{Re}\left[\frac{1}{j\omega} |\mathbf{X}| e^{j\phi} e^{j\omega t}\right] \Leftrightarrow \frac{1}{j\omega} \mathbf{X}$$

CORRESPONDS TO



# Summary of Cosine-reference Phasors

$$\sin(\theta) = \cos\left(\theta - \frac{\pi}{2}\right)$$

| $x(t)$                                  |                       | $\mathbf{X}$                   |
|-----------------------------------------|-----------------------|--------------------------------|
| $A \cos \omega t$                       | $\longleftrightarrow$ | $A$                            |
| $A \cos(\omega t + \phi)$               | $\longleftrightarrow$ | $Ae^{j\phi}$                   |
| $-A \cos(\omega t + \phi)$              | $\longleftrightarrow$ | $Ae^{j(\phi \pm \pi)}$         |
| $A \sin \omega t$                       | $\longleftrightarrow$ | $Ae^{-j\pi/2} = -jA$           |
| $A \sin(\omega t + \phi)$               | $\longleftrightarrow$ | $Ae^{j(\phi - \pi/2)}$         |
| $-A \sin(\omega t + \phi)$              | $\longleftrightarrow$ | $Ae^{j(\phi + \pi/2)}$         |
| $\frac{d}{dt}[A \cos(\omega t + \phi)]$ | $\longleftrightarrow$ | $j\omega Ae^{j\phi}$           |
| $\int A \cos(\omega t' + \phi) dt'$     | $\longleftrightarrow$ | $\frac{1}{j\omega} Ae^{j\phi}$ |



# Example

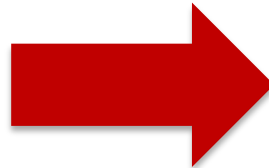
Calculate the output for the following system given the input  $x(t)$ , using phasors

$$\frac{d^2 y}{dt^2} + a \frac{dy}{dt} + by = x(t)$$

$$a = 300$$

$$b = 5 \times 10^4$$

$$x(t) = 10 \sin(100t + 60^\circ)$$



CONVERT TO PHASOR NOTATION

$$\begin{aligned} x(t) &= 10 \sin(100t + 60^\circ) \\ &= 10 \cos(100t + 60^\circ - 90^\circ) \\ &= 10 \cos(100t - 30^\circ) \end{aligned}$$

$$\Rightarrow \mathbf{X} = 10e^{-j30^\circ}$$

THE FREQUENCY  $\omega = 100$  IS IMPLICIT



## Example (contd)

---

Other phasors we need:

$$\frac{d^2 y}{dt^2} + a \frac{dy}{dt} + by = x(t)$$

$$y(t) \rightarrow \mathbf{Y}$$

$$\frac{dy}{dt} \rightarrow j\omega \mathbf{Y}$$

$$\frac{d^2 y}{dt^2} \rightarrow (j\omega)^2 \mathbf{Y}$$

**SIMILAR TO THE LAPLACE TRANSFORM**

**No Initial conditions to worry about since phasors represent the steady state**



## Example (contd)

The differential equation is transformed as follows:

$$\frac{d^2 y}{dt^2} + a \frac{dy}{dt} + by = x(t)$$

Remember:

THE FREQUENCY  $\omega = 100$  IS IMPLICIT

$$(j\omega)^2 \mathbf{Y} + aj\omega \mathbf{Y} + b\mathbf{Y} = \mathbf{X} = 10e^{-j30^\circ}$$

$$\Rightarrow \mathbf{Y} = \frac{10e^{-j30^\circ}}{-\omega^2 + ja\omega + b} = \frac{10e^{-j30^\circ}}{-100^2 + j300 \times 100 + 5 \times 10^4}$$

$$= \frac{10e^{-j30^\circ}}{10^4(4 + j3)} = \frac{10e^{-j30^\circ}}{10^4 \times 5e^{j36.87^\circ}}$$

$$= 2 \times 10^{-4} e^{j(-30^\circ - 36.87^\circ)} = 2 \times 10^{-4} e^{-j66.87^\circ} \Rightarrow y(t) = 2 \times 10^{-4} \cos(100t - 66.87^\circ)$$



# Summary

---

**We learned how to build a circuit given a transfer function**

**Basic lessons:**

- 1. Avoid differentiators, and use only integrators**
- 2. Direct-form circuits offer ways to save on op-amps compared to a naïve implementation from partial fraction expansions**



# For the Next Class

---

- **Study Sections 5-1 – 5-2 (new chapter)**

## **How to Study:**

1. **Read the assigned sections to understand the basic concepts, and aim to memorize the concepts**
2. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
3. **Write down your questions and bring them to class**